



AI and Cyberspace: Foundations for Safe Artificial General Intelligence

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Abstract

This paper surveys the intellectual foundations that inform Hanzo AI’s approach to artificial general intelligence safety. Drawing on work by Bostrom, Yudkowsky, Harris, Russell, Drexler, and others, we organize the field into four pillars: the nature and trajectory of machine intelligence, the risk landscape, technical alignment, and the governance of cyberspace as the substrate on which AGI will operate. We argue that AI safety and cybersecurity are deeply entangled—AGI risk cannot be addressed without securing the computational infrastructure it inhabits—and outline Hanzo’s position that building safe AGI requires simultaneous progress across ethics, technical alignment, cybersecurity, and social coordination.

1 Introduction

*“A great design appears at first insane;
But chance will soon seem quaint and
blind, And such an exemplary think-
ing brain Will soon by thinkers be de-
signed.” — Goethe, Faust*

Artificial intelligence has crossed from theoretical curiosity to practical capability. Language models generate coherent text. Vision systems exceed human performance on specific benchmarks. Reinforcement learning agents master games that were considered decades from solution. The question is no longer whether machines will match human cognitive capability, but when—and whether the transition will be safe.

This paper organizes the foundational literature on AI safety and cyberspace governance into a coherent framework, and describes how it informs the design principles of Hanzo AI.

2 Intelligence: Definitions and Trajectory

Shane Legg catalogued 71 definitions of intelligence, converging on the concept of optimization power divided by resources used [1]. Bostrom’s *Superintelligence* [2] explores takeoff scenarios: a sufficiently advanced AI that can improve its own architecture may trigger an intelligence explosion whose speed—hard or soft—determines how much time humanity has to prepare.

Yudkowsky’s Intelligence Explosion Microeconomics [3] provides a quantitative framework: if each generation of self-improving AI produces a successor that is δ more capable, and each improvement cycle takes time τ , the trajectory is exponential when δ/τ exceeds a critical threshold. The Foom Debate between Yudkowsky and Hanson [4] crystallizes the central disagreement: whether takeoff will be fast enough to preclude human intervention.

3 The Risk Landscape

3.1 Orthogonality and Instrumental Convergence

Bostrom’s Orthogonality Thesis [5] establishes that intelligence and goals are independent: a superintelligent system could pursue any objective, including ones catastrophic to humanity. Omohundro’s Basic AI Drives [6] shows that sufficiently advanced systems of *any* design will converge on instrumental subgoals—acquiring resources, self-preservation, goal-content integrity—that are dangerous even if the terminal goal is benign.

3.2 Malicious Use

Brundage et al. [7] document near-term risks from adversarial applications of existing AI: automated spear-phishing, deepfake generation, vulnerability discovery, and autonomous weapons. These risks materialize long before AGI.

3.3 Speed of Response

Yudkowsky’s “There Is No Fire Alarm for AI Safety” [8] argues that there will be no unambiguous warning sign before a critical capability threshold is crossed. Preparation must therefore be prospective, not reactive.

4 Technical Alignment

The alignment problem is the challenge of ensuring that an AI system’s objectives match human values. Key approaches:

- **Coherent Extrapolated Volition (CEV).** Yudkowsky’s proposal [9] to define AI goals as the values humanity *would* converge on given unlimited reflection time.
- **Iterated Distillation and Amplification.** Christiano’s [10] approach to training AI by recursively decomposing complex value judgments into simpler ones that humans can verify.
- **Cooperative Inverse Reinforcement Learning.** Russell’s framework [11] in which the AI is uncertain about the human’s objective and actively seeks to learn it through interaction.
- **Intelligence Distillation.** Drexler’s reframing [12] of the problem in terms of comprehensive AI services rather than autonomous agents.

5 Cyberspace: The Substrate

“Governments of the Industrial World, you weary giants of flesh and steel, I come from Cyberspace, the new home of Mind.” — John Perry Barlow, A Declaration of the Independence of Cyberspace

AGI will not exist in a vacuum. It will inhabit cyberspace—the global computational infrastructure comprising networks, data centers, edge devices, and satellites. The security properties of this substrate directly constrain the safety properties of any AI system running on it.

5.1 Cybersecurity as Existential Infrastructure

Peterson, Miller, and Duettmann [13] argue that cybersecurity is a near-term, undervalued catastrophic risk more imminent than AGI risk itself. Yampolskiy [14] proposes applying cybersecurity principles—defense in depth, principle of least privilege—directly to AI safety. Christiano [15] notes that alignment and security are inseparable: an aligned AI on a compromised system is as dangerous as a misaligned AI on a secure one.

5.2 Computational Governance

Bratton’s *The Stack* [16] describes planetary-scale computation as an accidental megastructure that functions simultaneously as infrastructure and governance. Drexler and Miller’s Agoric Papers [17] treat computation as an economic system—markets for CPU cycles, memory, and

bandwidth. These frameworks inform the design of decentralized AI infrastructure that resists single points of failure and centralized control.

5.3 Quantum Computing

The arrival of practical quantum computing [18] introduces both opportunity (faster optimization, novel ML architectures) and threat (cryptographic assumption invalidation). Systems designed for AGI safety must be post-quantum from the ground up.

6 Social Coordination

Technical alignment is necessary but insufficient. Bostrom et al. [19] identify policy desiderata for superintelligence development. Dafoe’s [20] reading guide maps the global politics of AI development. Armstrong, Bostrom, and Schulman [21] model the race dynamics that push developers toward unsafe speed over careful alignment.

Sandberg, Bostrom, and Douglas’s Unilateralist’s Curse [22] formalizes the observation that in a group of actors, the one most inclined to act unilaterally is also most likely to be wrong about the consequences. This argues for coordination mechanisms that prevent any single actor from deploying transformative AI without broad consensus.

7 Hanzo’s Position

Hanzo AI’s approach to safety rests on four commitments:

1. **Ethics first.** Value alignment is a prerequisite, not an afterthought. Systems are designed to augment human agency, not supplant it.
2. **Secure substrate.** AI infrastructure must be built on cybersecurity foundations—hardware attestation, post-quantum cryptography, decentralized trust—before capability scaling.
3. **Open coordination.** Safety research is shared openly. Competitive advantage comes from engineering execution, not from hoarding safety insights.
4. **Prospective preparation.** There is no fire alarm. Build safety into the architecture now, not after capability thresholds are crossed.

8 Conclusion

The path to artificial general intelligence passes through cyberspace. The safety of AGI systems depends not only on technical alignment but on the security and governance of the computational infrastructure they inhabit. By grounding its work in the interdisciplinary literature spanning AI risk, technical alignment, cybersecurity, and social coordination, Hanzo AI builds toward a future in which transformative AI benefits all of humanity.

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